Edge-Based Defense in Networks

A Modification of Bloch, Chatterjee & Dutta (2023)

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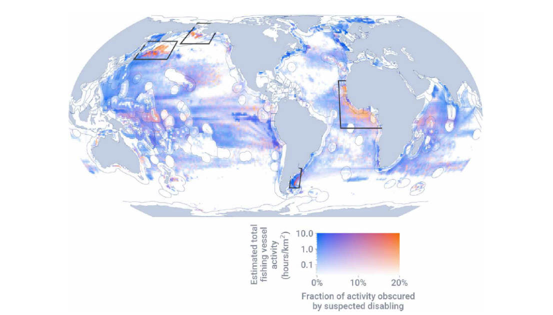
Strategic adversaries on networks



Estimated global fishing activity. Orange marks suspected illegal activity — West Africa is one of three concern zones.

Welch et al., *Science Advances* (2022) • Global Fishing Watch

The problem.

Illegal fishing reaches places no single country owns. Vessels move along sea routes between national waters and high seas, taking what isn't theirs to take.

The adversary is strategic.

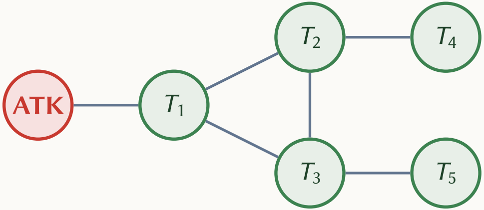
Routes are chosen, not random. The weakest gap is the one used.

The defenders are many.

Neighbouring states watch the same waters. Defense is naturally shared.

What is a network?

A network is a graph $G = (V, E)$: nodes V are locations, edges E are direct connections between them.



Nodes.

Locations the attacker can reach.

Edges.

The connections between locations.

The attacker.

A single adversary outside the network.

The defenders.

One at each target. They invest to intercept.

Two traditions, one gap

	Node Defense		Edge Defense
	Centralized	Continuous	Centralized
Authors	Dziubiński & Goyal Cerdeiro et al.	**Bloch, Chatterjee & Dutta (2023)**	Washburn & Wood Mavronicolas et al.
Investment	Binary	Continuous	Discrete or budgeted
Defenders	Single planner	Many	Single planner

The gap — *Continuous, decentralized investment — placed on the connections.*

The Bloch baseline: setup

$$\alpha_{k(p)}(p) = \prod_{\substack{j \in V(p) \\ j \neq 0,\, j \neq k(p)}} (1 - x_j)$$

$$U(\pi, x) = \sum_{p \in P} \pi(p)\, \beta_{k(p)}(p)\, b_{k(p)}$$

$$V_i(\pi, x) = -\sum_{\substack{p \in P \\ k(p) = i}} \pi(p)\, \beta_i(p)\, b_i - \frac{x_i^2}{2}$$

Path survival.

The attacker survives the interior of the path if it slips past every defender between the attacker node and the target.

Attacker's expected payoff.

Sum over every path, weighted by how often the attacker uses it, times the chance of getting through, times the target's value.

Defender's payoff.

Expected loss from attacks on its own target, minus the cost of its investment. Defenders are the only players who pay a cost.

The Bloch baseline: equilibrium

Bloch distinguishes two kinds of target. A first target is reached directly from the attacker node; a deeper target is reached only by passing through other targets that are themselves attacked.

$$x_i = 1 - \frac{U}{b_i}, \quad q_i = \frac{1}{b_i}\left(1 - \frac{U}{b_i}\right)$$

$$x_i = 1 - \frac{b_k}{b_i}, \quad q_i = \frac{b_k}{b_i\, U}\left(1 - \frac{b_k}{b_i}\right)$$

First-target equilibrium.

Higher-value targets invest more and are attacked more often. U is the common payoff across all attacked targets.

Deeper-target equilibrium.

For a target i reached only through its predecessor k. The predecessor's value b_k replaces U as the relevant threshold.

What Bloch leaves on the table

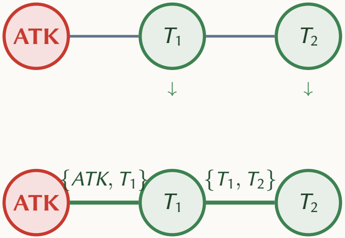
Defense lives at the location.

In Bloch, every defender's investment sits at its own node. The attacker is caught *at* a target, not on the way to one.

But corridors are shared.

A shipping lane, a forest road, a fishing zone — none of these belong to one party alone. Real defense happens *on the way*, by whoever watches that stretch.

Bloch: defense at nodes



Edge-based: defense on edges

The modification, stated

Move investment from nodes to edges. Every edge has two defenders.

Bloch's baseline

$$S(\text{node } i) \quad = \quad 1 - x_i$$

One defender per node. Survival is whatever it fails to catch.

The modification

$$S(e) \quad = \quad (1 - x_{a,e})(1 - x_{b,e})$$

Two defenders per edge. The attacker escapes only if *both* sides fail.

Cost is split: each defender i pays $\sum_{e \ni i} x_{i,e}^2/2$, summed over its incident edges.

What carries over from Bloch

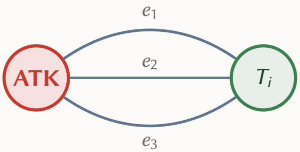
Three of the four key lemmas in Bloch's analysis depend only on the attacker's optimization. They survive the move to edges.

Equal payoff across the support	Every attacked target gives the attacker the same payoff U.
Optimal predecessor has lowest value	If a target can be reached through several support members, the attacker passes through the one with the lowest value.
Values increase along attack paths	Target values rise as the attacker moves deeper. Survival decay is offset by rising value.

Plus: equilibrium exists and is unique, under the same genericity assumption Bloch uses.

What changes (1): the multi-path factor

A target reachable through several free-highway edges is structurally weaker: each edge must be defended separately.



$m_i = 3$ free-highway edges reaching the same target

$$\text{Bloch:} \quad q_i = \frac{b_i - U}{b_i^2}$$

$$\text{Edge-based:} \quad q_i = \frac{m_i\,(b_i - U)}{b_i^2}$$

The factor m_i counts the free-highway edges feeding the target.

More edges $\Rightarrow$ more attack probability $\Rightarrow$ weaker target.

What changes (2): strategic substitutes

When two defenders share an edge, one's investment changes the other's incentive.

$$x^*_{i,e} \quad = \quad q\, b_i\, (1 \;-\; x_{j,e})$$

$$\frac{\partial x^*_{i,e}}{\partial x_{j,e}} \quad = \quad -q\, b_i \quad < \quad 0$$

If my neighbor invests more, I invest less.

The slope is strictly negative whenever attacks flow through the edge. Defenders' efforts substitute for each other.

But the edge cannot stay undefended.

If the other side invests zero, my best response is strictly positive. $(0, 0)$ is never an equilibrium when attacks reach the edge.

Other structural changes

Three further differences from Bloch's framework, briefly.

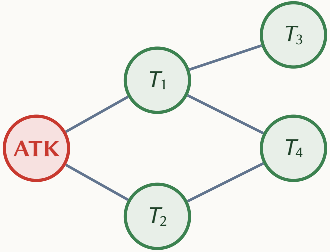
Mutual coverage on edges	When a target has m_i free-highway edges, the attacker spreads attack equally across them, and the defender invests equally on each.
No Braess paradox	Adding a connection in Bloch can hurt the attacker. Here, every new free-highway edge strictly raises attacker payoff U. Braess has no room to operate.
Mixed configurations	A target can be reached by both free-highway edges *and* through a predecessor. Each edge gets its own equilibrium equation. No analog in Bloch.

Managing a shared edge: four possibilities

Two endpoints, two modelling choices. Crossing them gives four cases.

	Independent interception $S(e) = (1-x_{i,e})(1-x_{j,e})$	**Joint interception** $S(e) = 1 - x_{i,e}\, x_{j,e}$
Separable cost	**Case A** (this chapter) Substitutes. Edge cannot stay undefended.	**Case C** Complements. $(0, 0)$ can be an equilibrium.
Shared cost	**Case B**	**Case D**

Example: a target with two paths



Values: b_1=0.20, b_2=0.25, b_3=0.50, b_4=0.70

T_4 has $m_4 = 2$ (reached via T_1 *and* T_2)

	Bloch	Edge-based
Attacker utility U	0.4020	**0.4773 (+18.7%)**
Attack on T_4 (q_4)	0.6081	**0.9091**
Attack on T_3 (q_3)	0.3920	0.0909
Investment x_3	0.1960	0.0455

More edges to defend $\Rightarrow$ higher U $\Rightarrow$ T_3 pushed to the margin $\Rightarrow$ attacks concentrate on T_4.

Cooperative defense: cheaper on a line

Centralized defender. Line network with values b_1=0.40, b_2=0.50, b_3=0.70.
T_2 is dropped from the support and used as a shield in both formulations.

	Node defense	**Edge-based**
Attacker utility U	0.2400	0.2400
Total defense cost	0.1396	**0.1142**
Cost saving	—	**18.2%**

Same attacker harm, lower cost. Splitting interception between two payers — each doing less — costs less

Beyond complete information

Real players rarely observe everything. The edge-based model extends in two natural directions.

Private target values.

Each defender knows its own value b_i, drawn from a common prior F. The attacker operates under the prior alone.

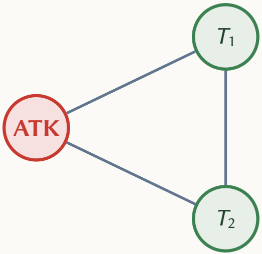
Partial network observation.

Each player observes its own incident edges with certainty, and assigns probability θ to every unseen edge.

Motivated by cybersecurity: Chang et al. (2024) show attackers target firms by observable signals (size, sales) — not by the true value at stake. The two extensions map onto these two information asymmetries.

Equilibrium concept: Bayes-Nash. Worked examples: the triangle network.

Private values on the triangle



Prior: F uniform on $(0.2, 0.8)$, $\mu = 0.5$.

info info	Complete Incomplete	
Attacker utility U	0.3869	**0.3726**
Shared-edge defended by	T_2 only	**both**
x on $\{T_1, T_2\}$ by T_1	0.0000	0.0168
x on $\{T_1, T_2\}$ by T_2	0.0392	0.0175

The headline: uncertainty disrupts perfect coordination. Both defenders cover the shared edge — neither knows for sure they are the lower-value target. Attacker is worse off.